

Field comparison of four positioning methods

FlexTDOA, Native DS-TWR, Passive DS-TWR and GPS RTK

Static outdoor data, common RTK-derived ENU frame, no UWB filtering

Test date: August 3, 2026, 16:58–17:14 EEST
Platform: 5 ESP32-S3 + DW3000 + PX1105R-01 modules, Raspberry Pi 5
Firmware: 8fce48f-dirty; repository HEAD 0c922dd
Configuration: T1 tag; A2, A3, A4, A5 anchors
Acquisition: three consecutive 300 s protocol blocks; GNSS configured at 1 Hz
Environment: outdoor, static geometry, no temporal filtering

Preliminary result. Under this single static field geometry, Native DS-TWR produced the tightest unfiltered position cloud: **1.39 cm CEP50** and **2.99 cm P95**. Passive DS-TWR delivered **26.12 complete positions/s**, approximately 11.4 times the Native rate, with **6.57 cm CEP50** and **10.06 cm P95**. The embedded FlexTDOA path published **54.68 positions/s**, while the common complete-slot reconstruction yielded 16.97 positions/s. Relative to the selected RTK reference, the three UWB RMSE values were 13.88 cm, 11.52 cm, and 10.15 cm, respectively.

RTK integrity warning. A3 reported RTK fixed throughout most of the campaign but changed by **2.62 m** between protocol blocks. The report therefore uses the only RTK block whose complete six-pair anchor geometry agrees with the static UWB anchor-range consensus. RTK is a valuable common reference here, but not an independently surveyed ground truth.

Reproducible preliminary technical report

Contents

1	Executive summary	1
2	System and tested protocol settings	1
2.1	Hardware and radio configuration	1
2.2	Position observables	2
3	Acquisition and common analysis method	2
3.1	Data blocks	2
3.2	Common ENU frame and accuracy reference	2
3.3	Frame assembly	3
4	RTK integrity audit	3
5	Position results	5
5.1	Accuracy relative to RTK	5
5.2	Static precision	6
6	Rate, delivery, and end-to-end output	8
7	Anchor-range consistency	9
8	Conclusions	9
9	Limitations	10
10	Reproducibility	10

1 Executive summary

This campaign is the first common data set containing all three implemented UWB protocols and GPS RTK. Each UWB protocol ran for five minutes in the same static outdoor layout. GNSS remained active on all modules, nominally at 1 Hz. The comparison deliberately separates two questions:

1. **Accuracy against the RTK reference:** horizontal distance between a UWB solution and the robust RTK-fixed T1 position.
2. **Precision:** radial dispersion about each method’s own robust median, which does not reward or penalize a constant offset.

All three UWB raw streams were reconstructed with the same unweighted, unfiltered two-dimensional offline solver and the same anchor coordinates. This avoids comparing different dashboard filters or embedded solver policies. The ESP32-published streams are listed separately as end-to-end results.

Table 1: Primary common-solver results. Accuracy is relative to the selected RTK reference; precision is dispersion about each method’s own median.

Method	N	Hz	Accuracy RMSE	Accuracy P95	Precision CEP50	Precision P95
FlexTDOA	5,093	16.97	13.88 cm	23.15 cm	9.23 cm	19.64 cm
Native DS-TWR	688	2.29	11.52 cm	13.50 cm	1.39 cm	2.99 cm
Passive DS-TWR	7,838	26.12	10.15 cm	14.97 cm	6.57 cm	10.06 cm
GPS RTK	544	0.60 observed	reference	reference	0.50 cm	1.10 cm

The result is nuanced. Native DS-TWR is clearly the most repeatable UWB method in this static trial. Passive DS-TWR happens to have the lowest RMSE against the chosen RTK position, but its wider cloud shows that this is partly a favorable cancellation of systematic bias; it is not more precise than Native DS-TWR. FlexTDOA remains the fastest end-to-end implementation, while Passive DS-TWR is the fastest source of complete frames reconstructed by the common solver in this particular capture.

2 System and tested protocol settings

2.1 Hardware and radio configuration

The system contains one receive-capable tag (T1) and four anchors (A2–A5). Every node combines an ESP32-S3, a Qorvo DW3000 UWB transceiver, and a SkyTraq PX1105R-01 GNSS receiver. All UWB methods used channel 9. GNSS RTK corrections were distributed through Wi-Fi. The tag and anchors were not moved between the three acquisition blocks.

Table 2: Runtime parameters captured in the metadata.

Method	Frame/slot	Relevant settings
FlexTDOA	3 ms slot, 12 ms four-anchor frame	250 μ s request subslot; 1,000 μ s request processing; 250 μ s response subslot and processing; 250 μ s guard time
Native DS-TWR	100 ms per anchor; 410 ms nominal four-anchor frame	10 ms frame gap; 90 ms RX timeout; 20 ms response delay; 20 ms final delay; native POLL–RESP–FINAL exchange
Passive DS-TWR	5 ms exchange slot; 1 ms gap	Robust Rotating schedule; 4 ms RX timeout; 1,000 μ s response; 1,000 μ s final delay; receive-only tag
GPS RTK	configured 1 Hz	PX1105R-01, NTRIP corrections through Wi-Fi; fixed solutions retained

2.2 Position observables

Native DS-TWR measures four absolute tag-to-anchor distances. The position is obtained by unweighted 2-D trilateration. FlexTDOA and Passive DS-TWR produce range-difference observations of the form

$$z_{ij} \simeq \|\mathbf{p} - \mathbf{a}_j\| - \|\mathbf{p} - \mathbf{a}_i\|, \quad (1)$$

and the common solver minimizes

$$\min_{\mathbf{p}} \sum_{(i,j)} (\|\mathbf{p} - \mathbf{a}_j\| - \|\mathbf{p} - \mathbf{a}_i\| - z_{ij})^2. \quad (2)$$

No median filter, moving average, Kalman filter, or trajectory smoothing was applied to any UWB position in the primary comparison. Physical corrections inside the radio measurements are not treated as temporal filtering.

3 Acquisition and common analysis method

3.1 Data blocks

The sequence was FlexTDOA, Native DS-TWR, then Passive DS-TWR. Each acquisition lasted 300 s after a 15 s warm-up. The collector polled module status at 50 Hz and requested GNSS at 1 Hz. Data were deduplicated by their native protocol keys before analysis.

Table 3: Captured data volume after key-based deduplication.

Block	UWB observations	Anchor ranges	ESP32 positions	GNSS events
FlexTDOA	192,787	203,581	16,408	924
Native DS-TWR	2,887 ranges	176	0	937
Passive DS-TWR	37,320	44,728	7,839	932

The absence of Native embedded positions is intentional: during this capture the Native position solver was running on the Raspberry Pi. Its raw ranges are therefore reconstructed offline by the common solver.

3.2 Common ENU frame and accuracy reference

RTK-fixed WGS84 latitude, longitude, and altitude were converted to ECEF and then to a local East–North–Up frame whose origin is the robust A2 position. The robust T1 median across all blocks is the horizontal accuracy reference. The tag changed by only 1.5 cm between block centers, so pooling its fixed samples is appropriate for this static test.

The anchor geometry could not be pooled blindly: A3 entered mutually inconsistent RTK-fixed solutions. Section 4 documents the selection of the coherent geometry. The resulting geometry is shown below.

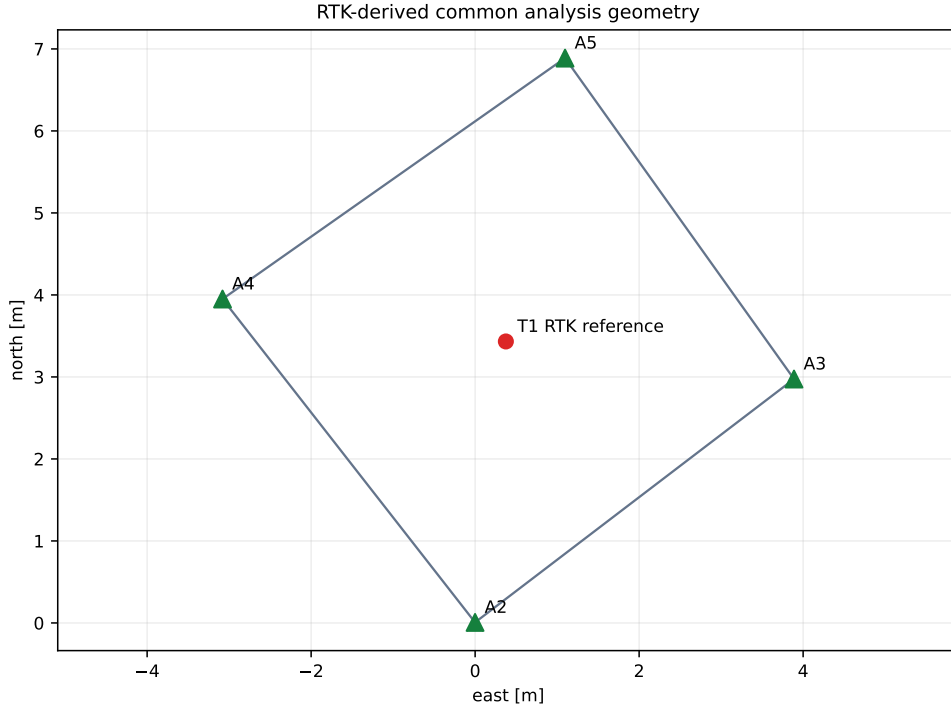


Figure 1: Common analysis geometry derived from the coherent RTK block. The T1 marker is the robust RTK-fixed reference position.

3.3 Frame assembly

Native DS-TWR uses a complete set only when all four ranges carry the same reconstructed cycle. FlexTDOA uses three responder observations belonging to one native slot. Passive DS-TWR emits the three responder exchanges in three consecutive slots; the common frame identifier is recovered from the ESP32 slot identifier and responder index. No Raspberry Pi wall-clock synchronization is used to number UWB frames.

4 RTK integrity audit

The GNSS receiver reported RTK fixed for more than 96% of the retained samples on every node. Nevertheless, solution status alone was insufficient: the A3 block center moved by 2.62 m while A2, A4, A5, and T1 each remained within 1.8 cm across the campaign.

Table 4: Maximum three-dimensional displacement between consecutive RTK block centers.

Module	T1	A2	A3	A4	A5
Maximum span	1.47 cm	1.21 cm	2.62 m	1.70 cm	1.48 cm

Because the physical anchor layout was static, the six robust anchor-to-anchor UWB medians provide a cross-sensor integrity check. For each RTK epoch, the six RTK 3-D separations were compared with the corresponding UWB consensus.

Table 5: RTK anchor-geometry integrity against the six-pair static UWB consensus. The selected geometry is bold.

RTK block	Complete pairs	Pair RMSE	Maximum residual
FlexTDOA block	6	1.068 m	2.232 m
Native DS-TWR block	6	0.848 m	1.757 m
Passive DS-TWR block	6	0.071 m	0.137 m

The Passive block is the only complete RTK geometry compatible with the radio distances and is used for all three UWB solvers. This procedure prevents a known false-fixed coordinate from generating metre-scale fictitious UWB errors. It also creates a limited dependence between the geometry reference and UWB, so this report must not be interpreted as a metrological calibration against independent survey equipment.

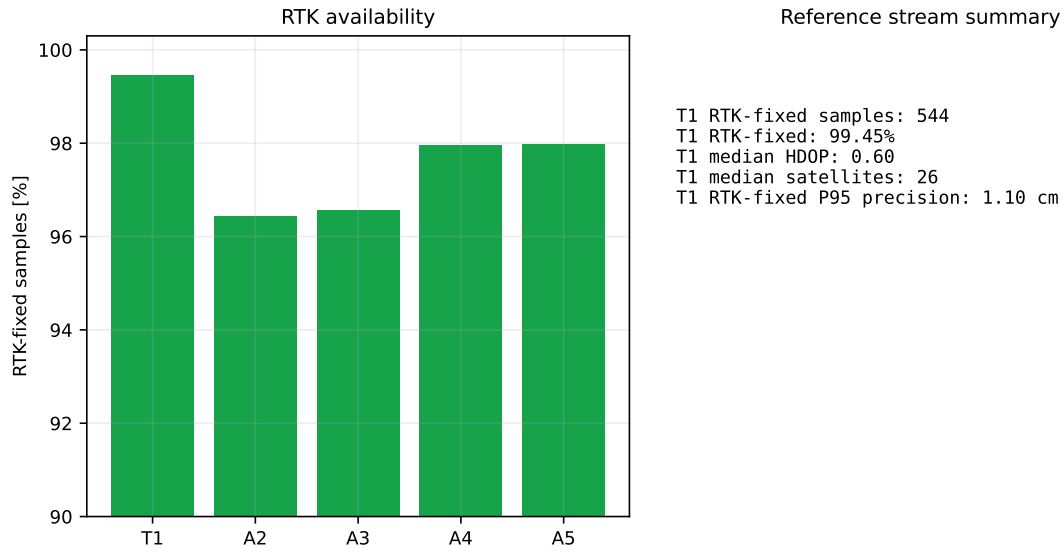


Figure 2: RTK status availability and T1 reference-stream statistics. The high fixed percentage does not reveal the A3 integrity failure.

5 Position results

5.1 Accuracy relative to RTK

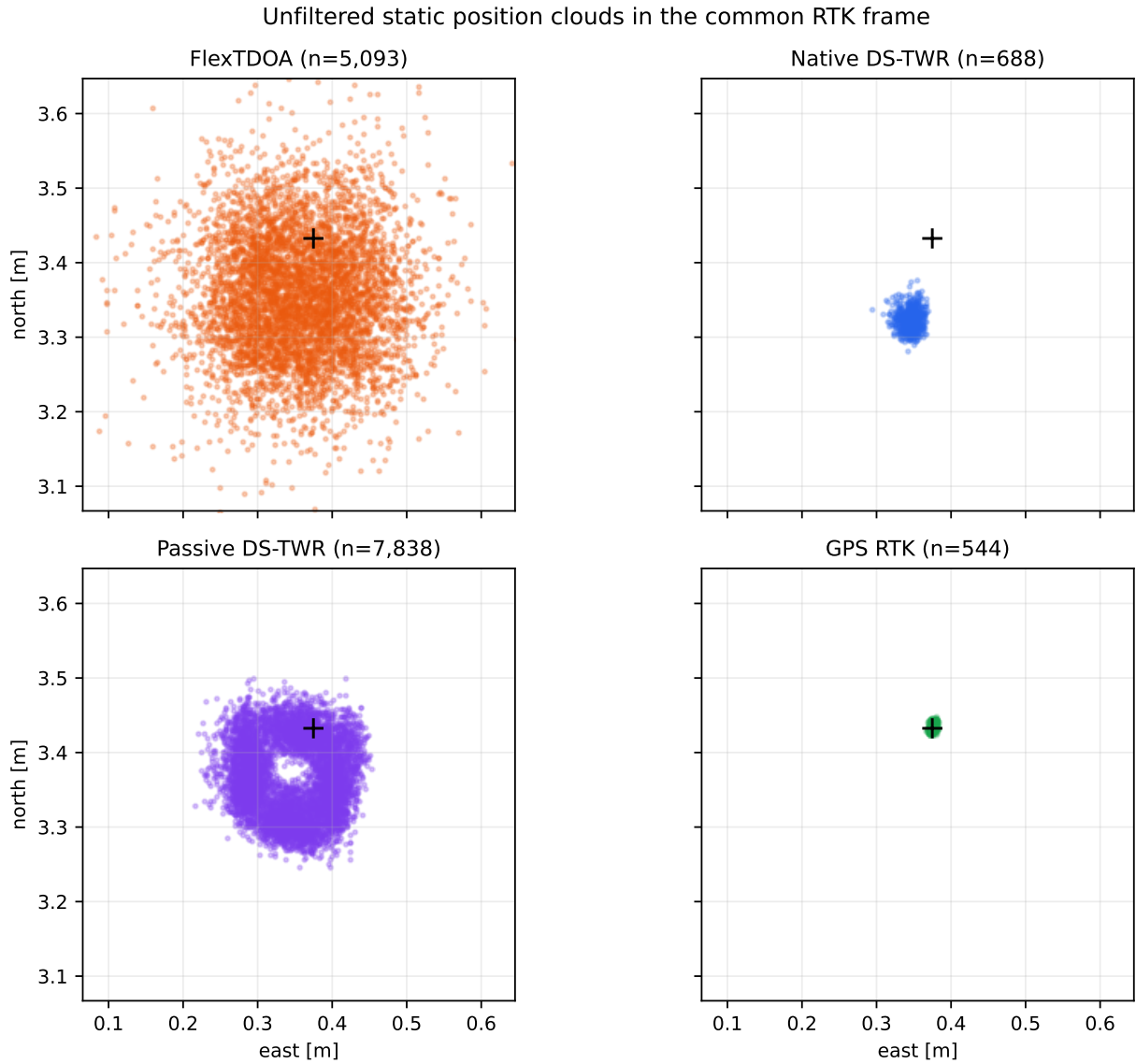


Figure 3: Unfiltered position clouds in the same RTK-derived coordinate frame. The black cross is the RTK-fixed T1 median. Identical axes make the different dispersions directly visible.

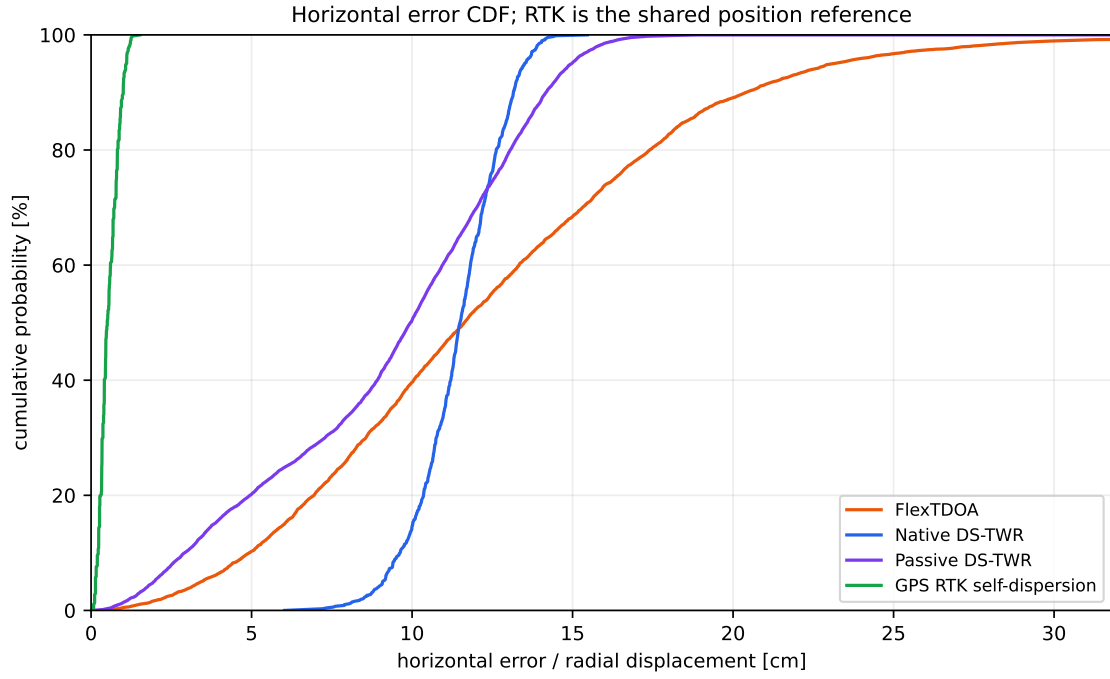


Figure 4: CDF of horizontal UWB error relative to T1 RTK. The green curve is RTK self-dispersion, not an independent RTK accuracy measurement.

The Passive DS-TWR median lies closer to T1 RTK than the Native DS-TWR median, which explains its slightly lower RMSE. Native DS-TWR remains much tighter: its complete error distribution spans only 10.0 cm from the observed minimum to maximum, whereas Passive contains a 51.4 cm maximum error. Therefore the accuracy ranking at this one point must be read together with precision and range bias, not from RMSE alone.

5.2 Static precision

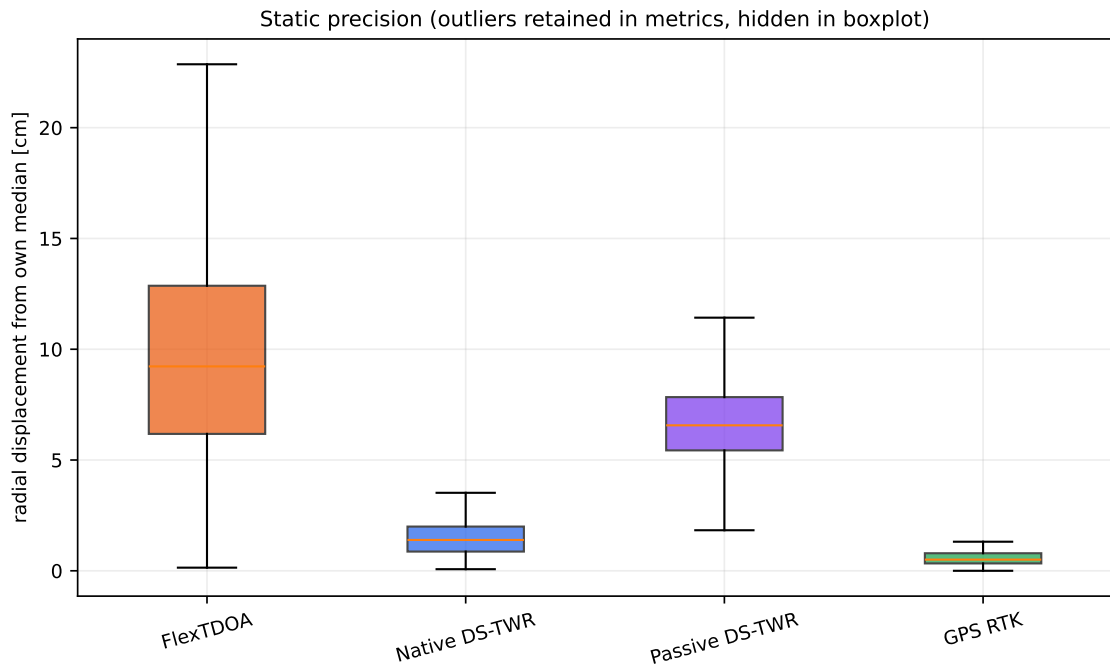


Figure 5: Radial displacement from each method's own median. Plot outliers are hidden for readability but retained in all tabulated metrics.

Native DS-TWR is the clear static-precision winner: 1.39 cm CEP50 and 2.99 cm P95. GPS RTK is approximately three times tighter at P95, while Passive DS-TWR and FlexTDOA are respectively 3.4 and 6.6 times wider than Native at P95.

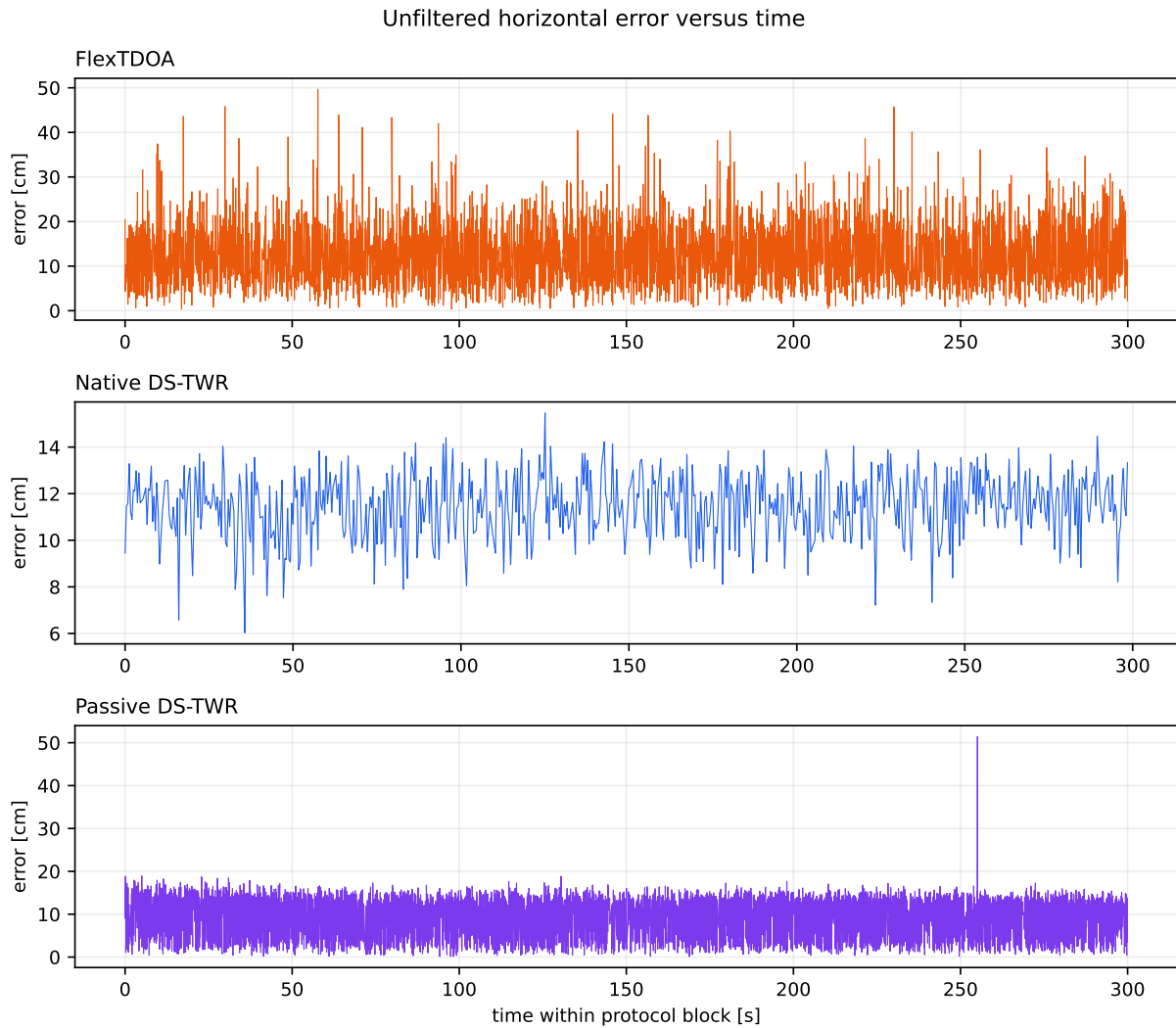


Figure 6: Horizontal UWB error versus time. No temporal filtering or ground-truth-based rejection was applied.

6 Rate, delivery, and end-to-end output

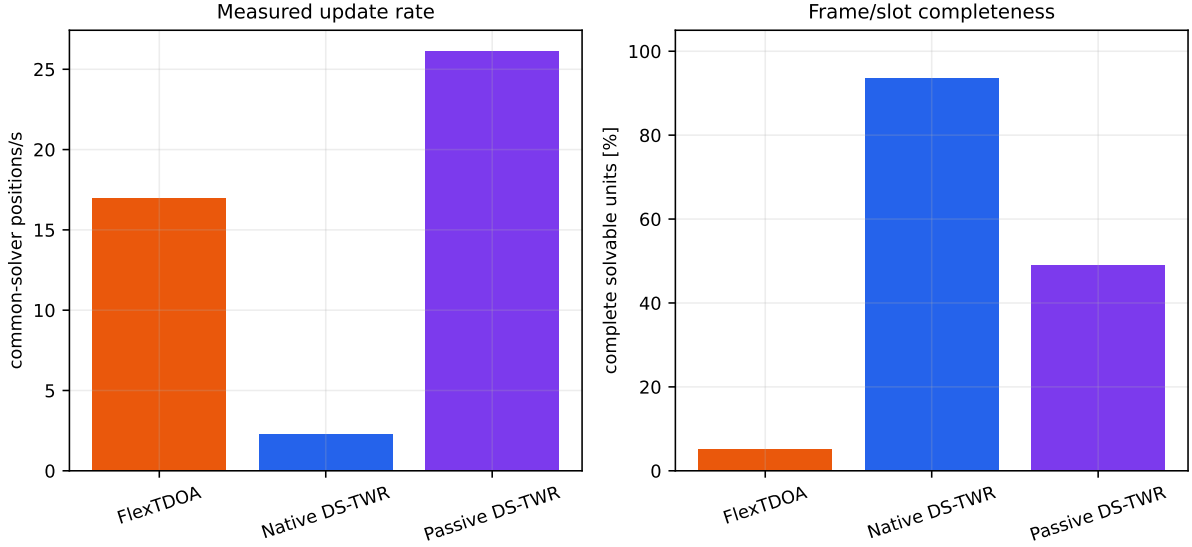


Figure 7: Common-solver update rate and completeness. Candidate-unit definitions are protocol-specific, so completeness should be interpreted as a delivery diagnostic rather than a direct efficiency ranking.

Table 6: Frame assembly and delivery diagnostics.

Method	Candidates	Complete	Complete	Median interval	P95 interval
FlexTDOA	97,543	5,093	5.22%	48.3 ms	193.9 ms
Native DS-TWR	735	688	93.61%	410.3 ms	811.2 ms
Passive DS-TWR	16,013	7,838	48.95%	29.9 ms	134.0 ms

FlexTDOA’s primary common-solver row deliberately requires all three responses from one physical slot. The product’s embedded solver publishes more often by using the current valid observation state. This produces the following end-to-end precision result, which is useful operationally but is not identical to the complete-frame comparison above.

Table 7: Positions published by the running ESP32 solver. Native used the Raspberry solver during this capture and therefore has no embedded row.

Method	N	Published Hz	Precision CEP50	Precision P95
FlexTDOA embedded	16,408	54.68	3.44 cm	7.51 cm
Passive DS-TWR embedded	7,839	26.13	5.06 cm	8.94 cm

This distinction resolves an apparent contradiction: FlexTDOA is the fastest live implementation, but Passive DS-TWR produced more fully coherent raw frames for the strict common solver in this capture.

7 Anchor-range consistency

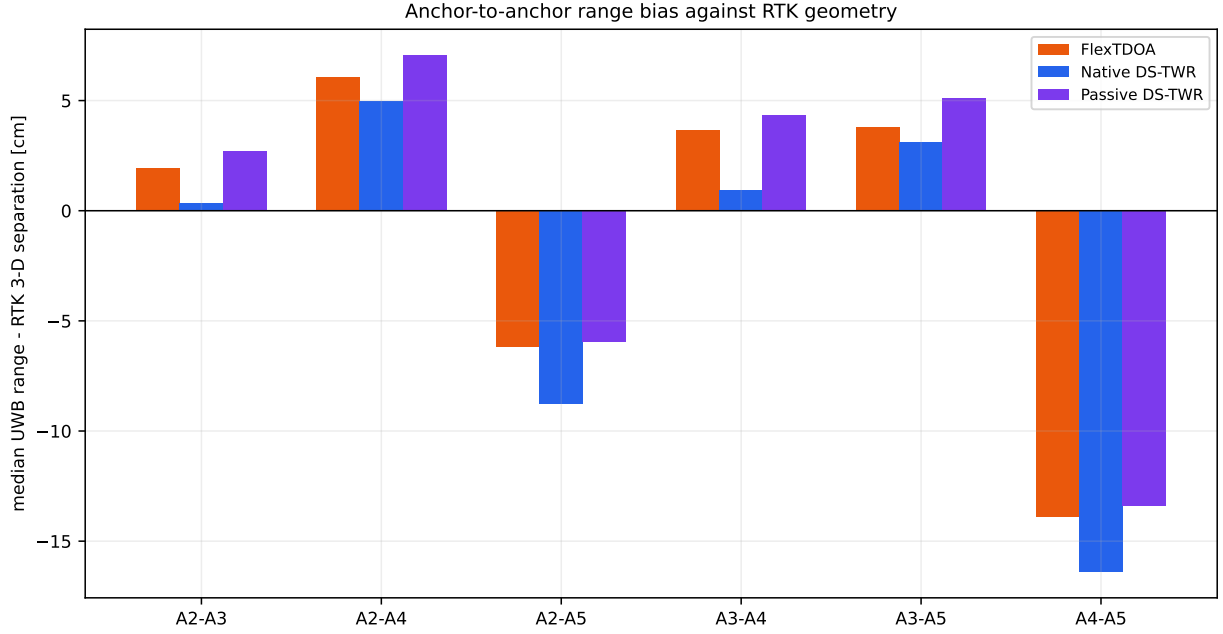


Figure 8: Median UWB anchor range minus the coherent RTK 3-D separation. These biases include both UWB hardware bias and residual RTK-geometry error.

All three protocols show the same sign pattern on most pairs, suggesting that a substantial component is hardware/path dependent rather than solver noise. The largest common discrepancy is A4–A5, approximately -13.4 to -16.4 cm. Native DS-TWR has the smallest range dispersion on most pairs, generally 1.3–3.1 cm standard deviation; Passive DS-TWR is similar on five pairs, while FlexTDOA anchor-range telemetry is wider at about 4.6–5.8 cm.

This pair structure is a clear calibration opportunity. A future independent survey should estimate antenna-delay and pair biases without using RTK epochs that fail the geometry integrity check.

8 Conclusions

1. **Native DS-TWR is the strongest static precision reference.** It achieves 2.99 cm P95 with no temporal filtering, but only 2.29 complete positions/s under the conservative 410 ms four-anchor profile.
2. **Passive DS-TWR is the best complete-frame speed/precision compromise in this capture.** It reaches 26.12 positions/s, 6.57 cm CEP50, and 10.06 cm P95 while the tag remains receive-only.
3. **FlexTDOA remains the fastest live output.** The ESP32 publishes 54.68 positions/s with 7.51 cm P95 self-dispersion. Its strict, complete-slot common reconstruction is slower and noisier because only 5.22% of candidate slots contain every required observation in the captured stream.
4. **Accuracy and precision lead to different rankings.** Passive DS-TWR has the lowest RMSE against this RTK reference, but Native DS-TWR has by far the smallest cloud. A single favorable bias cancellation must not be mistaken for general accuracy superiority.
5. **RTK fixed is not an integrity guarantee.** A3 produced a 2.62 m false-fixed displacement. Cross-checking static inter-anchor geometry is mandatory before RTK is used as an accuracy reference.

Recommended next experiment. Repeat randomized five-minute blocks at several tag positions and include a dynamic trajectory. Survey anchor coordinates independently or use a verified local reference, and randomize protocol order to separate protocol behavior from time-varying RF and GNSS conditions. Retain both common complete-frame and actual end-to-end results.

9 Limitations

This report describes one static geometry and three sequential five-minute blocks. It does not test edge geometry, NLOS, antenna orientation, speed, or multi-tag scaling. The protocol order was not randomized. GNSS was configured at 1 Hz, but the distinct retained T1 fixed stream was observed at 0.60 Hz over the full 900 s campaign. Most importantly, RTK is part of the tested system and the coherent anchor epoch was selected using static UWB pair consistency; the reported UWB “accuracy” is therefore a practical cross-method comparison, not traceable absolute accuracy.

10 Reproducibility

The report directory contains:

- raw JSONL captures and per-block metadata under `data/`;
- `analysis_summary.json`, `position_metrics.csv`, `anchor_pair_metrics.csv`, and `rtk_geometry_integrity.csv`;
- all common-solver positions in `common_solver_positions.csv`;
- vector and raster figures under `figures/`;
- checksums in `SHA256SUMS`.

The deterministic analysis command is:

```
python3 tools/uwb_gps_rtk_report.py \  
    reports/uwb_gps_rtk_comparison_20260803
```